



Published in final edited form as:

J Addict Med. 2019 ; 13(5): 385–395. doi:10.1097/ADM.0000000000000512.

Medical Burden of Disease Among Individuals in Recovery from Alcohol and Other Drug Problems in the United States: Findings from the National Recovery Survey

David Eddie, Ph.D.¹, M. Claire Greene, Ph.D., M.P.H.², William L. White, M.A.³, John F. Kelly, Ph.D.¹

¹ Recovery Research Institute, Center for Addiction Medicine, Massachusetts General Hospital, Harvard Medical School

² Department of Psychiatry, Columbia University Medical Center & New York State Psychiatric Institute

³ Chestnut Health Systems

Abstract

Objectives: The medical sequelae of alcohol and other drug (AOD) problems exact a prodigious personal and societal cost, but little is known about the specific prevalence of such medical problems and their relationship to quality of life and indices of well-being among those recovering from problematic AOD use. To better characterize the lifetime physical disease burden, this study investigated the prevalence of medical conditions commonly caused or exacerbated by excessive and chronic AOD exposure in a nationally representative sample of U.S. adults in AOD problem recovery. Comparisons were made to the general U.S. population. Demographic and clinical correlates of disease prevalence were also investigated along with the relationship between distinct medical conditions and indices of quality of life/well-being.

Methods: Cross-sectional nationally-representative survey of the U.S. adult population who report resolving an AOD problem ($n=2002$). Weighted lifetime prevalence of common medical conditions were estimated and compared to the U.S. population. Demographic and clinical correlates of medical conditions, as well as overall disease burden, were estimated using logistic regression.

Results: Relative to the general population, prevalence of hepatitis C, chronic obstructive pulmonary disease, heart disease and diabetes were elevated. Likelihood of having a lifetime diagnosis of a specific disease was related to primary substance used and sex. Quality of life was lower among those with physical disease histories relative to those without.

Conclusions: Findings highlight the increased medical burden associated with AOD problems, and speak to the need for earlier and more sustained intervention for AOD problems, greater integration of addiction treatment and primary health care, and longitudinal research to explore the complex, dynamic relationships between AOD use and physical disease.

Keywords

Alcohol; drugs; physical disease; disease burden; recovery; addiction; medical

Introduction

The personal, social, and economic costs of the medical sequelae of alcohol and other drug (AOD) problems are enormous (Whiteford et al. 2013) with direct and indirect effects of AOD exposure and attendant lifestyles giving rise to an array of medical conditions that can reduce life expectancy, and increase years lived with disease and disability (Hall and Degenhardt 2009, Whiteford et al. 2013). Further, the medical sequelae of AOD problems typically persist after individuals enter recovery, such that medical and psychiatric problems may continue to affect individuals' quality of life long after an AOD problem is resolved (Di Sclafani et al. 2007, Fein et al. 2008). Studies of clinical populations have found increases in health-related quality of life among people in recovery from a substance use disorder (Garner et al. 2014). However, community surveys are lacking on the health status of people who have successfully overcome an AOD problem that could illuminate the prevalence of common medical conditions in this population, and how living with such medical conditions may affect quality of life.

In order to better characterize the lifetime disease burden of problematic AOD use, the present study investigated the prevalence of medical conditions commonly caused or exacerbated by chronic AOD exposure in a nationally representative sample of U.S. adults in recovery from an AOD problem (Kelly et al. 2017). Where possible, in order to provide comparisons with the general population, observed frequencies of medical conditions in our recovering AOD sample were compared to national averages using the Centers for Disease Control Prevention (CDC), National Health and Nutrition Examination Survey (NHANES) (Centers for Disease Control Prevention 2018b) dataset. Also, because of the different kinds of medical problems that can arise from exposure to various substances (e.g., alcohol vs cocaine etc.), disease prevalence was also examined across these different substances. Finally, demographic and clinical correlates of disease prevalence were investigated and the relationship between distinct medical conditions and quality of life was explored.

Methods

Procedure

The National Recovery Survey (NRS) targeted the U.S. noninstitutionalized civilian population 18 years or older that had resolved an AOD problem, as indicated by affirmative response to the screener question: "Did you used to have a problem with drugs or alcohol, but no longer do?" (Kelly et al. 2017). Data were collected by the survey company GfK, using a probability sampling approach to select respondents at random. GfK screened everyone in their "KnowledgePanel" (GfK 2013), which consists of approximately 55,000 adult individuals aged 18 and older. The KnowledgePanel uses address-based sampling (ABS) to randomly select individuals from 97% of all U.S. households based on the U.S. Postal Service's Delivery Sequence File. If necessary, GfK provides individuals with a web-

enabled computer and free internet service. Using this ABS approach, GfK is able to include households that a) have unlisted telephone numbers, b) do not have landline telephones, c) are cell phone only, d) do not have current internet access, and e) do not have devices to access the internet. This type of broad scale sampling helps redress socioeconomic differences in landline telephone use and internet access. For the current study, a representative subset of 39,809 individuals from the GfK KnowledgePanel received the screening question. In order to draw this subsample, GfK uses a probability proportional to size (PPS) sampling approach, a patented strategy (U.S. Patent No. 7,269,570) unique to GfK. PPS assures that subsamples from a finite panel membership remains a reliable approximation of the entire U.S. Population. See www.gfk.com/fileadmin/user_upload/dyna_content/US/documents/KnowledgePanel_A_Methodological_Overview.pdf for more information on GfK's probability-based sampling methodology.

The survey was first piloted on 20 individuals over 3 days in July 2016 then formally administered over 19 days in July-August 2016. Median time to completion was 24 minutes (*IQR* = 18–36 minutes). Of those in the initial sampling frame (*N*= 39,809), 25,229 individuals responded to the screening question (63.4%). This response rate is comparable to most other current nationally representative surveys, including the National Epidemiologic Survey on Alcohol and Related Conditions-III (NESARC-III; 60.1%; Grant et al. 2015) the 2015 National Survey on Drug Use and Health (NSDUH; 58.3%; Center for Behavioral Health Statistics and Quality 2016) and the 2013–2014 National Health and Nutrition Examination Survey (NHANES; 68.5%; Centers for Disease Control and Prevention 2013). Data were weighted to accurately represent the civilian population using the method of “iterative proportional fitting” (Battaglia et al. 2009).

In order to produce unbiased estimates of population parameters from these respondents, GfK first computed *base weights* then made post-stratification adjustments according to benchmarks from the Current Population Survey (CPS), conducted March 2015 by the U.S. Bureau of the Census, along eight dimensions: 1) sex; 2) age; 3) race/Hispanic ethnicity; 4) education; 5) geographical region; 6) household income; 7) home ownership status; and 8) metropolitan area. A sample of 2,002 individuals were included in the analyses. GfK's population-based probability sampling approach has been vetted and validated in dozens of published studies in the medical and behavioral health fields (https://www.gfk.com/fileadmin/user_upload/dyna_content/US/documents/GfK_Bibliography.pdf).

KnowledgePanel-derived estimates are comparable to those derived from national surveys that used non-internet methodologies to recruit and collect data (Bethell et al. 2004, Novak et al. 2007, Heeren et al. 2008, Chang and Krosnick 2009, Yeager et al. 2011). Heeren et al. (2008) for example, showed that estimates of current drinking obtained through a GfK KnowledgePanel-derived sample were similar to those obtained by NESARC. The final weighted sample was *n*= 2,002.

Measures

Demographic characteristics—Demographic data were derived both from GfK's existing KnowledgePanel data (collected prior to the survey) as well as from survey data. Regarding previously collected demographic data, participants reported the following: 1)

age, 2) level of education (less than high school, high school, some college, bachelor's degree or higher), 3) race/ethnicity (White/non-Hispanic, Black/non-Hispanic, Other/non-Hispanic, 2+ races/non-Hispanic, Hispanic), 4) sex (male, female), 5) household income (19 categories ranging from less than \$5,000 to \$175,000 or more), and 6) current employment.

Physical disease history—Participants were asked whether they had ever been told by a healthcare provider if they had one or more of nine medical conditions (Dennis et al. 2008): 1) alcohol-related liver disease, 2) hepatitis C, 3) tuberculosis (TB), 4) the human immunodeficiency virus (HIV), or acquired immune deficiency syndrome (AIDS), 5) another sexually transmitted infection (e.g., gonorrhea), 6) chronic obstructive pulmonary disease, 7) cancer, 8) heart disease, or 9) diabetes.

Quality of life—Quality of life was assessed using the EUROHIS-QOL (Schmidt et al. 2005), a widely used eight-item measure of quality of life, adapted from the World Health Organization Quality of Life – Brief Version (WHOQOL-BREF). Item responses are on Likert scales from 1 to 5 (e.g., “How satisfied are you with your personal relationships?” 1 = *very dissatisfied* to 5 = *very satisfied*). The measure has strong psychometric properties, including good to excellent predictive validity (i.e., significant discrimination between individuals with and without a health condition), convergent validity with other measures of health and wellness ($r = 0.4$ to 0.6), and internal consistency ($\alpha = .83$). Its internal consistency was excellent in the current sample ($\alpha = .90$).

Statistical Analysis—The weighted lifetime prevalence of each of the nine medical conditions was calculated in the full sample and by primary substance used (alcohol, cannabis, opioids, stimulants, and other drugs). For diseases where 2015–2016 National Health and Nutrition Examination Survey (NHANES) data were available, we compared prevalence estimates in the full sample to corresponding weighted prevalence estimates in the general population of non-institutionalized resident adults in the United States captured by the NHANES. We used direct standardization to account for differences in the age distribution in these two samples. Because the NHANES 2015–2016 did not include self-report items inquiring about whether participants had been told in their lifetime by a health professional that they had tuberculosis, HIV/AIDS, or sexually transmitted infections, we were unable to compare these estimates between samples. Additionally, there were discrepancies between the NRS and NHANES surveys in how they asked about liver disease and cancer. The NRS survey specifically asked whether participants had been told by a health professional that they had alcohol-related liver disease, while the NHANES asked about any type of liver disease. Similarly, the NRS survey asked if participants had been told they had cancer, while the NHANES sample was asked whether they had been told they had cancer *or* a malignancy.

We examined demographic and clinical correlates of each of these medical conditions as well as overall disease burden (1 vs. 0 medical conditions, 2+ vs. 0 medical conditions) using unadjusted logistic regression models. Demographic correlates included age (in years), sex, education, race/ethnicity, household income, marital status, and employment. Clinical correlates included primary substance used, age of onset of regular use, number of substances used ten or more times, years since AOD problem resolution, number of years

smoked (lifetime), utilization of outpatient substance use treatment (yes/no), utilization of inpatient or residential substance use treatment (yes/no), mutual-help meeting attendance (e.g., Alcoholics Anonymous, SMART Recovery), happiness (Meyers and Smith 1995), self-esteem (Robins et al. 2001), quality of life (Schmidt et al. 2005), psychological distress (Kessler et al. 2003), and recovery capital (Vilsaint et al. 2017). All models incorporated sampling weights and were conducted in Stata version 14.

Results

Participant characteristics and their pathways to overcoming an AOD problem are displayed in Table 1.

Prevalence of different diseases among individuals who have resolved a problem with AOD

Thirty-seven percent of adults who have resolved an AOD problem reported being told by a health professional that they had one or more of the following health conditions during their lifetime: Alcohol-related liver disease (4.84%, 95% CI: 3.66, 6.39), hepatitis C (5.49%, 95% CI: 4.17, 7.19), tuberculosis (1.42%, 95% CI: 0.74, 2.70), HIV/AIDS (2.61%, 95% CI: 1.68, 4.04), a sexually transmitted infection (8.69%, 95% CI: 7.12, 10.57), chronic obstructive pulmonary disease (COPD; 4.99%, 95% CI: 4.03, 6.16), cancer (5.50%, 95% CI: 4.41, 6.83), heart disease (5.29%, 95% CI: 4.24, 6.58), or diabetes (13.62%, 95% CI: 11.77, 15.72)(Table 2). When compared to age-standardized national prevalence, hepatitis C, COPD, heart disease and diabetes were significantly more prevalent among adults who had resolved an AOD problem relative to the general non-institutionalized adult population in the U.S.

Prevalence of disease by primary substance used

Notable differences in the prevalence of disease was observed when comparing the primary substance type used by participants. The prevalence of alcohol-related liver disease was highest among participants who reported alcohol as their primary substance (6.65%); however, this elevated prevalence was only significantly greater than that observed for individuals who reported opioids or stimulants as their primary substance. The prevalence of hepatitis C was significantly higher in the opioid and stimulant groups relative to participants who reported alcohol as their primary substance. Lifetime prevalence of HIV/AIDS and sexually transmitted infections were significantly higher in the stimulant group relative to the alcohol group. The prevalence of cancer was not significantly different between primary substance groups. Heart disease was most common in the primary alcohol group, however, this difference was only statistically significantly different from the opioid group, which had the lowest prevalence of heart disease. Diabetes was least common among participants who reported cannabis as their primary substance, which was significantly less prevalent than that observed in the alcohol group. No significant differences in the prevalence of tuberculosis or COPD were observed between primary substance groups.

Demographic and clinical correlates of medical conditions and disease burden

Similar to epidemiologic patterns observed in general population samples, the prevalence of physical disease was related to various demographic characteristics. In general, younger age, higher education, being married or living with a partner, and being employed were associated with having no physical diseases. Specifically, for each additional year of age, there is a 1.03-fold and 1.07-fold increase in the odds of having 1 or at least 2 physical diseases relative to no physical diseases, respectively. Having at least a high school diploma was associated with approximately a 55–60% reduction in the odds of having a physical disease; however, the relationship between physical disease and level of education was not linear such that further education (e.g., some college, bachelor's degree or higher) was not associated with a significantly greater reduction in the odds of having one physical disease as compared to individuals with only a high school education. When comparing the odds of having two or more physical diseases versus none, the association with education level was negative and linear, but not significant. Demographic factors associated with reduced odds of having two or more physical diseases versus none included Hispanic ethnicity, female sex, a household income greater than \$50,000USD, and being employed.

With regard to specific physical conditions, older age was related to elevated odds of COPD, cancer, heart disease and diabetes. Participants with a bachelor's degree or higher displayed lower odds of COPD, hepatitis C, HIV/AIDS and diabetes relative to individuals without a high school diploma. Compared to males, females were less likely to report having had alcohol-related liver disease, hepatitis C, cancer, heart disease or diabetes, but more likely to have had a sexually transmitted infection. Lifetime history of hepatitis C was less common in individuals with a higher current household income, who were currently married or employed. A household income of \$50,000USD or greater was also associated with lower odds of having COPD or diabetes. Current employment was associated with lower odds of lifetime alcohol-related liver disease, HIV/AIDS, COPD, cancer, heart disease and diabetes.

Clinical factors were also found to relate to physical health conditions. Older age at first regular use of primary substance, more years since problem resolution, additional years having smoked tobacco (lifetime), lower quality of life, and having utilized inpatient treatment, outpatient treatment or mutual-help for a substance use problem were related to increased odds of reporting one or more physical diseases. More specifically, each additional substance used ten or more times in one's life, older age of onset in years, and additional years since problem resolution were associated with a 4–7% increase in the odds of having two or more physical diseases as compared to no physical diseases. Relative to participants who reported alcohol as their primary substance, those who reported cannabis as their primary substance displayed a 60% reduction in the odds of having two or more physical diseases as compared to none. Inpatient, outpatient, former mutual-help participation and past 3-month mutual-help participation were each associated with a 2.3-to 2.7-fold increase in the odds of having two or more physical diseases as compared to none. Past 3-month mutual-help meeting attendance was similarly associated with a 2.3-fold increase in the odds of having one physical disease relative to none.

Examination of specific physical diseases revealed variable relationships with clinical characteristics. For example, older age at first regular use of primary substance, additional

years having smoked (lifetime), and additional years since problem resolution were associated with increased odds of COPD, cancer and heart disease. Years since problem resolution and additional years having smoked (lifetime) were also associated with an increase in the odds of diabetes. Having ever participated in outpatient treatment and participating in a mutual-help meeting within the past three months was associated with increased odds of alcohol-related liver disease. Utilization of outpatient, inpatient or mutual-help meetings and additional years having smoked (lifetime) were associated with increased odds of hepatitis C. Increased years having smoked was also associated with increased odds of HIV/AIDS. Inpatient treatment and having formerly attended a mutual-help meeting, but not in the past three months was associated with an increase in the odds of diabetes.

Relationship between disease and quality of life

Having no lifetime history of physical health conditions was associated with better quality of life. Alcohol-related liver disease, COPD, and diabetes were each associated with significantly lower levels of quality of life. Alcohol-related liver disease and tuberculosis were also significantly associated with greater psychological distress. Conversely, having suffered from cancer at some point during one's life was significantly associated with better quality of life across all indicators included in this study (i.e., quality of life, happiness, self-esteem, psychological distress, recovery capital).

Discussion

The prodigious psychological, social, and interpersonal impact of excessive and chronic AOD use is widely appreciated and has been well characterized. Less well appreciated, however, is the *physical disease* burden arising from AOD problems. The present study is novel in that it documents for the first time national prevalence estimates of common physical diseases among individuals who have overcome significant AOD problems. The ability to examine the impact of certain demographic and clinical subgroups, as well as quantity of substances used (e.g., substances used 10 or more times) are the most novel aspects of this analysis compared to the extant literature. Findings indicate an overall higher prevalence of certain diseases relative to the general population, that certain demographic and clinical sub-groups within the AOD recovering population appear to be at variable risk of acquiring a number of different diseases, and that overall quality of life is lower among those who have suffered a physical disease relative to those who have not.

Taken together, findings are indicative of greater lifetime physical disease burden among adults who have resolved a problem with AOD relative to the general non-institutionalized adult, U.S. population. When compared to age-standardized national lifetime prevalence rates, hepatitis C, COPD, heart disease, and diabetes were significantly more prevalent in the present NRS sample compared to the general population. Further, as might be expected, participants endorsing diagnoses of physical diseases reported current poorer quality of life compared to those not reporting physical disease.

Direct, standardized, age-weighted comparisons between the present sample (NRS) and the general population (NHANES) were not possible for alcohol-related liver disease, tuberculosis, HIV/AIDS, sexually transmitted infections, or cancer as the NHANES survey

either does not assess for these diseases or assesses them in ways not commensurate with the NRS study.

Though direct comparisons were not possible for these diseases, we note that the present NRS sample experienced rates for these diseases at least comparable with previously reported estimates for the general population, and in some instances notably greater. For instance, the prevalence of self-reported lifetime sexually transmitted infections in the present NRS sample (8.69%) was greater among women in the sample, but homologous to NHANES bio-assay assessed sexually transmitted infection rates in the general population (7.42%). Similarly, the cancer prevalence rate in the NRS sample (5.50%) was comparable to estimated prevalence rates in the general population (4.80%; Bluethmann et al. 2016), but relative to the general population is about 15% higher ($5.5 - 4.8 = 0.7$; $0.7 \div 4.8 \times 100 = 15\%$). The incidence of HIV/AIDS, however, appears to be seven to eight times higher in the NRS sample (2.6%), compared to the general population (0.34%; Centers for Disease Control Prevention 2018a), though these CDC estimates are for the year 2015 (versus 2016 for the NRS sample) and include people aged 13 and older (versus 18 and older for the NRS sample). Regarding tuberculosis, 1.42% of this NRS sample endorsed being told by a healthcare professional that they have this disease. This is comparable to findings from the 2011–2012 NHANES for a positive response to a tuberculosis skin test (1.40%), but higher than the rate of NHANES participants endorsing being told by a healthcare provider that they have *active* tuberculosis (0.39%), and lower than NHANES participants having a positive tuberculosis blood test (5.82%). Because we cannot know if NRS participants were responding affirmatively to being told by a health care professional that they have tuberculosis based on a positive skin or blood test, or because they have active symptoms, we cannot draw inferences from this comparison. The lifetime prevalence rate of alcohol-related liver disease for the general population has not been previously reported, precluding comparisons for this class of disease based on other research. One would expect, however, that those endorsing alcohol as their primary substance, have higher rates of alcohol-related liver disease than the general population.

Individuals who primarily used classes of drugs commonly associated with injection such as opiates or stimulants had higher rates of diseases known to be transmitted by sharing needles, including hepatitis C and HIV. This observation aligns with a fairly large body of work showing elevated rates of hepatitis C and HIV in individuals who inject drugs (e.g., Strathdee et al. 2010, Nelson et al. 2011). Participants primarily using stimulant also had significantly higher rates of sexually transmitted infections than those primarily using alcohol, which supports previous findings reporting higher incidence of sexually transmitted infections among individuals who use stimulants (particularly methamphetamine) who are thought to engage in greater and/or more risky sexual activity than individuals who do not use stimulants (Molitor et al. 1998). Surprisingly, individuals endorsing cannabis as their primary substance used were not significantly different in terms of rates of alcohol-related liver disease compared to those endorsing alcohol as their primary substance used, which may be a function of greater alcohol use among those who for whom cannabis is also problematic. It is also possible that some individuals experienced problematic alcohol use earlier in life, and in an attempt to mitigate alcohol harms, shifted to primary cannabis use later. This would suggest the value in assessing early drinking histories and potential

alcohol-related medical conditions among people presently diagnosed with cannabis use disorder.

It is noteworthy that history of outpatient and inpatient treatment, mutual-help meeting attendance, and past 3-month mutual-help meeting attendance were associated with a large increase (~2-fold) in the odds of having physical disease relative to none. Given the majority of individuals with AOD problems never present for treatment, and those that do present for treatment typically have greater AOD problem severity, it is possible that outpatient and inpatient treatment, and mutual-help participation in our sample is a marker for greater addiction severity (Humphreys et al. 1998). Logically it follows that as a result of greater addiction severity, these individuals are also more likely to suffer from medical sequelae and physical diseases as a result of greater intensity and chronicity of AOD exposure. It is also possible that regardless of addiction severity, individuals with more physical disease burden are also suffering from greater psychological distress, which addiction treatment and mutual-help organization participation can help ameliorate (Kelly et al. 2009, Kelly et al. 2017).

Also of note, greater time since AOD problem resolution was associated with a higher probability of having a lifetime diagnosis of a number of diseases, including COPD, cancer, heart disease, and diabetes. This is a function of participant age. That is, individuals who have longer time since AOD problem resolution will tend to be older, and the natural increase in probability of disease occurrence with the passing of time. Supporting this postulate, greater age was also associated with a higher probability of having a lifetime diagnosis of the same diseases.

Not surprisingly, lifetime tobacco use was positively associated with likelihood of a number of diseases (i.e., hepatitis C, HIV/AIDS, COPD, cancer, heart disease, diabetes; Table 4), and disease burden (Table 5). Given the high rates of tobacco use in individuals with AOD problems compared to the general population (Weinberger et al. 2016), it is possible tobacco use, in and of itself, is contributing markedly to the disease burden of AOD problems. It is possible too that the combination of tobacco use and AOD problems produces synergistic health detriments (e.g., Taylor and Rehm 2006). The importance of this finding is underscored by studies concluding that people treated for alcohol or other drug use disorders are more likely to subsequently die from smoking-related diseases than from other drug-related causes (e.g., Hser et al. 1994, Hurt et al. 1996). Although, recent research suggests greater numbers of those in recovery from AOD problems are now quitting smoking and quitting sooner (Kelly et al. 2019), increased calls to integrate smoking cessation into addiction treatment programs remain underscored by research suggesting that smoking cessation may improve recovery outcomes for other drug dependencies (Prochaska et al. 2004).

Generally speaking, having one or more lifetime disease diagnoses was associated with lower current quality of life (Tables 4 & 5). Counterintuitively though, participants with cancer diagnosis histories endorsed, on average, greater quality of life, happiness, self-esteem, recovery capital, and lower psychological distress than those without. This may be a function of participants being in cancer remission at the time of surveying, and having a more positive life outlook as a result of surviving cancer—a phenomenon previously

reported in the cancer literature (e.g., Bower et al. 2005). Based on the present data it cannot be known if individuals who have resolved a problem with AOD and have active cancer also endorse greater quality of life and positive affect.

Although the cross-sectional nature of this study precludes causal inferences being made regarding AOD exposure and increased prevalence of physical diseases, existing research on the mechanisms through which AOD exposure can increase the risk of certain physical diseases (e.g., Mathers et al. 2008, Rehm et al. 2009, Nelson et al. 2011), suggests a high probability that the increased prevalence of certain physical diseases in this NRS sample, relative to the general population, is either directly or indirectly related to AOD problem prevalence. This in turn, can affect the long-term quality of life, functioning, and psychological well-being of those who are able to achieve recovery. Given the enormous number of individuals affected by AOD problems (Kelly et al. 2017), the findings reported here underscore the public health need for early AOD interventions and more aggressive and sustained AOD care to prevent the occurrence of serious illnesses that are associated with increased health care utilization and costs, as well as long-term diminished quality of life. The findings also suggest the need for increased integration between addiction treatment and primary health care as well as the potential role of the primary care physician in performing ongoing recovery checkups and assistance in managing chronic health conditions for patients recovering from AOD problems.

Limitations

The present findings should be viewed in light of limitations pertaining to the study design: 1) These data are cross-sectional and reflect lifetime occurrence of disease. It is possible that some participants may have developed these diseases before developing a problem with AOD. 2) There is no way to know whether these conditions resolved or improved after individuals' AOD problems were resolved, and how present quality of life temporally aligns with participants' disease history. 3) Differential screening, access to medical services, and insurance status were not retrospectively assessed in this study, and could have contributed in unknown ways to observed differences in disease prevalence. 4) The NRS study's stem question ("Did you used to have a problem with drugs or alcohol but no longer do?") was designed to capture the broader population of individuals affected by AOD problems, including those who have not been formally diagnosed, or do not identify as having a substance use disorder history, but whose AOD problems nevertheless contribute to addiction disease burden. An inherent strength and limitation of this approach is that the specific parameters of what constitutes an AOD problem and overcoming an AOD problem are determined subjectively by the participants. 5) Although the total sample size was large, the prevalence of a number of conditions was quite low. As such, we may be underpowered to detect differences between some subgroups. On the other hand, this exploratory study tests multiple bivariate associations, thus increasing risk of type I error.

Conclusions and Future Directions

It is well documented that AOD problems are related to premature mortality, and an increased burden of disease, as measured by disability-adjusted life-years lost (DALYs; HHS 2016). However, little is known about the specific prevalence of such medical problems and

their relationship to quality of life and indices of well-being among those recovering from problematic AOD use. Using a nationally-representative study of U.S. adults who have resolved an AOD problem, this study highlights the elevated prevalence of certain diseases (e.g., hepatitis C, COPD, heart disease and diabetes) in this group compared to the U.S. adult general population. Further, the likelihood of having a lifetime diagnosis of a specific disease appears to be influenced by sex, and primary substance used. Finally, quality of life is lower among those who have suffered a physical disease relative to those who have not. Though some of these findings are intuitive (e.g., individuals who primarily used classes of drugs commonly associated with injection such as opiates or stimulants had higher rates of hepatitis C and HIV), others are less intuitive (e.g., individuals endorsing cannabis as their primary substance did not have significantly lower rates of alcohol-related liver disease compared to those for whom alcohol was primary). In combination with other epidemiological and clinical findings regarding the deleterious impact of AOD use on health, this study suggests the need for earlier and more assertive intervention to help mitigate potential AOD-induced disease burden, and underscores the call for better integration of medical and substance use disorder services. Also, further longitudinal research is needed that can help uncover the complex and dynamic relationships between AOD use and the onset of various physical diseases over time to better articulate the related trajectory of AOD problems and attendant medical conditions, and the frequency at which the medical sequelae of AOD problems resolve after individuals overcome their problems with AOD.

References

1. Battaglia MP, Izrael D, Hoaglin DC, Frankel MR. Practical considerations in raking survey data. *Survey Practice* 2009;2:1–10.
2. Bethell C, Fiorillo J, Lansky D, Hendryx M, Knickman J. Online consumer surveys as a methodology for assessing the quality of the United States health care system. *J Med Internet Res* 2004;6:e2. [PubMed: 15111268]
3. Bluethmann SM, Mariotto AB, Rowland JH. Anticipating the “Silver Tsunami”: prevalence trajectories and comorbidity burden among older cancer survivors in the United States. *Cancer Epidemiology, Biomarkers & Prevention* 2016;25:1029–1036.
4. Bower JE, Meyerowitz BE, Benaards CA, Rowland JH, Ganz PA, Desmond KA. Perceptions of positive meaning and vulnerability following breast cancer: Predictors and outcomes among long-term breast cancer survivors. *Ann Behav Med* 2005;29:236–245. [PubMed: 15946118]
5. Center for Behavioral Health Statistics and Quality. 2015 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions. In: Rockville, MD: 2016.
6. Centers for Disease Control and Prevention. Unweighted response rates for The National Health and Nutrition Examination Survey 2011–2012. Available at: https://www.cdc.gov/nchs/nhanes/response_rates_cps.htm 3 25, 2018.
7. Centers for Disease Control Prevention. HIV Surveillance Supplemental Report. In: Atlanta, GA, USA: Centers for Disease Control Prevention, 2018a:1–77.
8. Centers for Disease Control Prevention. National Health and Nutrition Examination Survey. In: Atlanta, GA, USA: Centers for Disease Control Prevention, 2018b.
9. Chang L, Krosnick JA. National surveys via RDD telephone interviewing versus the Internet: Comparing sample representativeness and response quality. *Public Opin Q* 2009;73:641–678.
10. Dennis DL, White M, Titus JC, Unsicker J. Global Appraisal of Individual Needs: Administration guide for the GAIN and related measures. In: Bloomington, IL: Chestnut Health Systems, 2008.

11. Di Sclafani V, Finn P, Fein G. Psychiatric comorbidity in long-term abstinent alcoholic individuals. *Alcohol Clinical and Experimental Research* 2007;31:795–803.
12. Fein G, Di Sclafani V, Finn P, Shumway R. Psychiatric comorbidity in older long-term abstinent alcoholics. *Addict Behav* 2008;33:1564–71. [PubMed: 18786773]
13. Garner BR, Scott CK, Dennis ML, Funk RR. The relationship between recovery and health-related quality of life. *J Subst Abuse Treat* 2014;47:293–298. [PubMed: 25012552]
14. GfK. GfK Knowledgepanel Design Summary. Available at: www.gfk.com/fileadmin/user_upload/dyna_content/US/documents/KnowledgePanel_A_Methodological_Overview.pdf. Accessed January 10, 2019.
15. Grant BF, Goldstein RB, Saha TD, et al. Epidemiology of DSM-5 alcohol use disorder: Results from the National Epidemiologic Survey on alcohol and related conditions III. *JAMA Psychiatry* 2015;72:757–766. [PubMed: 26039070]
16. Hall W, Degenhardt L. Adverse health effects of non-medical cannabis use. *The Lancet* 2009;374:1383–1391.
17. Heeren T, Edwards EM, Dennis JM, Rodkin S, Hingson RW, Rosenbloom DL. A comparison of results from an alcohol survey of a prerecruited Internet panel and the National Epidemiologic Survey on Alcohol and Related Conditions. *Alcoholism: Clinical and Experimental Research* 2008;32:222–229.
18. HHS. Recovery: The many paths to wellness In: Facing addiction in America: The Surgeon General's report on alcohol, drugs, and health. Washington, DC, USA: U.S. Department of Health and Human Services (HHS), Office of the Surgeon General, 2016.
19. Hser YI, McCarthy WJ, Anglin MD. Tobacco use as a distal predictor of mortality among long-term narcotics addicts. *Preventive Medicine* 1994;23:61–9. [PubMed: 8016035]
20. Humphreys K, Kaskutas LA, Weisner C. The Alcoholics Anonymous Affiliation Scale: development, reliability, and norms for diverse treated and untreated populations. *Alcoholism: Clinical and Experimental Research* 1998;22:974–8.
21. Hurt RD, Offord KP, Croghan IT, et al. Mortality following inpatient addictions treatment. Role of tobacco use in a community-based cohort. *Journal of the American Medical Association* 1996;275:1097–103. [PubMed: 8601929]
22. Kelly JF, Bergman B, Hoepfner BB, Vilsaint CL, White WL. Prevalence and pathways of recovery from drug and alcohol problems in the United States population: Implications for practice, research, and policy. *Drug Alcohol Depend* 2017;181:162–169. [PubMed: 29055821]
23. Kelly JF, Greene MC, Bergman B, Hoepfner B. Smoking cessation in the context of recovery from drug and alcohol problems: Prevalence, predictors, and cohort effects in a national U.S. sample. *Drug Alcohol Depend* 2019;195:6–12. [PubMed: 30557814]
24. Kelly JF, Magill M, Stout RL. How do people recover from alcohol dependence? A systematic review of the research on mechanisms of behavior change in Alcoholics Anonymous. *Addiction Research & Theory* 2009;17:236–259.
25. Kessler RC, Barker PR, Colpe LJ, et al. Screening for serious mental illness in the general population. *Arch Gen Psychiatry* 2003;60:184–189. [PubMed: 12578436]
26. Mathers BM, Degenhardt L, Phillips B, et al. Global epidemiology of injecting drug use and HIV among people who inject drugs: A systematic review. *The Lancet* 2008;372:1733–1745.
27. Meyers RJ, Smith JE. Clinical guide to alcohol treatment: The community reinforcement approach. New York, NY, USA: Guilford Press, 1995.
28. Molitor F, Truax SR, Ruiz JD, Sun RK. Association of methamphetamine use during sex with risky sexual behaviors and HIV infection among non-injection drug users. *Western Journal of Medicine* 1998;168:93. [PubMed: 9499742]
29. Nelson PK, Mathers BM, Cowie B, et al. Global epidemiology of hepatitis B and hepatitis C in people who inject drugs: results of systematic reviews. *The Lancet* 2011;378:571–583.
30. Novak SP, Kroutil LA, Williams RL, Van Brunt DL. The nonmedical use of prescription ADHD medications: results from a national Internet panel. *Substance Abuse Treatment, Prevention, and Policy* 2007;2:32.

31. Prochaska JJ, Delucchi K, Hall SM. A meta-analysis of smoking cessation interventions with individuals in substance abuse treatment or recovery. *J Consult Clin Psychol* 2004;72:1144–56. [PubMed: 15612860]
32. Rehm J, Mathers C, Popova S, Thavorncharoensap M, Teerawattananon Y, Patra J. Global burden of disease and injury and economic cost attributable to alcohol use and alcohol-use disorders. *The Lancet* 2009;373:2223–2233.
33. Robins RW, Hendin HM, Trzesniewski KH. Measuring global self-esteem: Construct validation of a single-item measure and the Rosenberg Self-Esteem Scale. *Personality and Social Psychology Bulletin* 2001;27:151–161.
34. Schmidt S, Mühlan H, Power M. The EUROHIS-QOL 8-item index: psychometric results of a cross-cultural field study. *The European Journal of Public Health* 2005;16:420–428. [PubMed: 16141303]
35. Strathdee SA, Hallett TB, Bobrova N, et al. HIV and risk environment for injecting drug users: the past, present, and future. *The Lancet* 2010;376:268–284.
36. Taylor B, Rehm J. When risk factors combine: the interaction between alcohol and smoking for aerodigestive cancer, coronary heart disease, and traffic and fire injury. *Addict Behav* 2006;31:1522–1535. [PubMed: 16443330]
37. Vilsaint CL, Kelly JF, Bergman BG, Groshkova T, Best D, White W. Development and validation of a Brief Assessment of Recovery Capital (BARC-10) for alcohol and drug use disorder. *Drug Alcohol Depend* 2017;177:71–76. [PubMed: 28578224]
38. Weinberger AH, Funk AP, Goodwin RD. A review of epidemiologic research on smoking behavior among persons with alcohol and illicit substance use disorders. *Preventive Medicine* 2016;92:148–159. [PubMed: 27196143]
39. Whiteford HA, Degenhardt L, Rehm J, et al. Global burden of disease attributable to mental and substance use disorders: Findings from the Global Burden of Disease Study 2010. *The Lancet* 2013;382:1575–1586.
40. Yeager DS, Krosnick JA, Chang L, et al. Comparing the accuracy of RDD telephone surveys and internet surveys conducted with probability and non-probability samples. *Public Opin Q* 2011;75:709–747.

Table 1.

Characteristics and pathways to recovery of U.S. adults who endorsed they "used to have a problem with drugs or alcohol, but no longer do" (9.1% [SE=0.28]; 22.35 million Americans). Adapted from Kelly et al., 2017.

Demographics	weighted %	SE
Sex		
Female	40.0	1.53
Male	60.0	1.53
Age		
18–24 years (emerging adulthood)	7.1	1.16
25–49 years (young adults)	45.2	1.63
50–64 years (mid-life stage adults, CDC)	34.7	1.43
65+ years (older adults)	13.0	0.76
Race & Ethnicity		
White, non-Hispanic	61.4	1.64
Black, non-Hispanic	13.8	1.19
Other, non-Hispanic	5.8	0.92
Hispanic	17.3	1.38
2+ races, non-Hispanic	1.7	0.30
Employment Status		
Working - as a paid employee	47.7	1.61
Working - self-employed	7.0	0.78
Not working - on temporary layoff from a job	1.5	0.50
Not working - looking for work	7.7	0.96
Not working - retired	12.0	0.80
Not working - disabled	15.6	1.14
Not working - other	8.5	0.93
Living Accommodations		
With family or other relatives	45.6	1.62
With group of friend(s) or non-family members (non-institutional)	3.6	0.73
Alone in own dwelling	29.7	1.37
Homeless	0.9	0.35
Hospital, rehabilitation facility, nursing home	0.2	0.16
Sober living environment (e.g., halfway house, Oxford House, sober dorm, etc.)	0.5	0.29
Other (please specify):	19.2	1.22
Clinical variables		
Time since problem resolution (in years) did not indicate	1.1	0.33
0–5 years	34.4	1.61
5–15 years	35.2	1.54
15+ years	29.3	1.32
Number of substances used 10+ times did not specify any substance	0.5	0.25
1 substance	26.8	1.41
2 substances	23.1	1.40

Demographics	weighted %	SE
3+ substances	49.6	1.61
Age of onset of first substance did not answer	2.0	0.48
< 15 years of age	47.1	1.61
> 15 years of age	50.9	1.61
Age of onset of problem substance did not identify any problem substance	12.6	1.14
< 15 years of age	34.5	1.54
> 15 years of age	52.9	1.61
Primary substance † did not identify any problem substance	12.6	1.14
Alcohol	51.2	1.61
Cannabis (e.g., marijuana, hashish)	11.0	1.13
Cocaine (e.g., coke, crack, freebase)	10.0	0.92
Methamphetamine (crank, meth, crystal)	7.3	0.90
Opioids (e.g., heroin, unprescribed fentanyl, methadone)	5.3	0.77
Other	2.7	0.50
Lifetime mental health disorder diagnoses		
Alcohol/other drug use disorder	16.8	1.18
Anxiety disorder	22.2	1.27
Mood disorder	16.9	1.13
Other disorder	10.9	0.97
Unsure if diagnosed	8.9	1.01
Never been diagnosed	48.2	1.61
Did not complete item	0.6	0.28
History of involvement in a drug court		
Never been arrested	49.0	1.60
Arrested, but no drug court	42.7	1.59
Arrested, and participated in a drug court	7.8	1.01
Did not complete item	0.5	0.27
Alcohol and other drug problem recovery pathways		
Professionally assisted recovery support (aka formal treatment)		
Outpatient addiction treatment	16.8	1.21
Inpatient or residential treatment	15.0	1.08
Alcohol/drug detoxification services	9.1	0.91
Did not use any of the above	68.9	1.49
Anti-relapse/craving medication use (any)	8.6	0.90
Alcohol		
Antabuse (Disulfiram)	2.4	0.4
Selincro (Nalmefene)	0.9	0.4
Revia (Naltrexone)	0.6	0.3
Campral (Acamprosate)	0.5	0.2
Topamax (Topiramate)	0.5	0.3
Lioresal (Baclofen)	0.2	0.2
Other	1.0	0.3

Demographics	weighted %	SE
Opioid		
Methadone	1.4	0.4
Orlaam (Levomethadyl acetate)	0.5	0.3
Suboxone (Buprenorphine-naloxone)	2.3	0.5
Subutex (Buprenorphine)	1.0	0.4
Revia (Oral naltrexone)	0.2	0.2
Vivitrol (Long-acting injectable naltrexone)	0.4	0.3
Other	0.4	0.2
Did not answer	0.6	0.3
Recovery support services		
Faith-based recovery services	9.2	0.94
Sober living environment	8.5	0.95
Recovery community centers	6.2	0.85
State or local recovery community organization	3.0	0.61
College recovery programs/communities	1.7	0.52
Recovery high schools	0.8	0.37
Did not use any of the above	78.2	1.33
Mutual-help organizations		
Alcoholics Anonymous (AA)	34.6	1.49
Narcotics Anonymous (NA)	17.5	1.23
Cocaine Anonymous (CA)	2.3	0.43
Celebrate Recovery	2.2	0.44
SMART Recovery	1.3	0.35
Women for Sobriety	1.2	0.37
Crystal Methamphetamine Anonymous (CMA)	0.8	0.37
Marijuana Anonymous (MA)	0.9	0.43
LifeRing Secular Recovery	0.4	0.27
Moderation Management	0.2	0.10
Secular Organizations for Sobriety (S.O.S.)	0.2	0.10
Other	3.2	0.47
Never attended a mutual-help meeting	54.9	1.60
Neither treatment programs, medication, services or mutual-help meetings	45.8	1.60

Table 2.

Prevalence (%) of medical conditions among those who have resolved and alcohol or other drug problem and adults in the general population with 95% confidence interval

Medical Condition	NRS Weighted Prevalence	NRS Weighted, Age-Standardized Prevalence	NHANES Weighted, Age-Standardized Prevalence
Alcohol-related liver disease	4.84 (3.66, 6.39)	—	—
Hepatitis C	5.49 (4.17, 7.19)	5.17 (4.16, 6.19)	1.15 (1.15, 1.16)
Tuberculosis	1.42 (0.74, 2.70)	—	—
Human immunodeficiency virus (HIV) / Acquired immune deficiency syndrome (AIDS)	2.61 (1.68, 4.04)	—	—
Sexually transmitted infections (STI)	8.69 (7.12, 10.57)	—	—
Chronic obstructive pulmonary disease (COPD)	4.99 (4.03, 6.16)	5.53 (4.30, 6.77)	3.12 (3.11, 3.12)
Cancer	5.50 (4.41, 6.83)	—	—
Heart disease	5.29 (4.24, 6.58)	6.19 (4.83, 7.55)	3.50 (3.50, 3.50)
Diabetes	13.62 (11.77, 15.72)	13.69 (11.90, 15.47)	10.73 (10.73, 10.74)

Notes. NRS= National Recovery Survey, NHANES= National Health and Nutrition Examination Survey

Table 3.

Demographic correlates of physical disease, odds ratios with 95% confidence intervals

	Alcohol-related liver disease	Hepatitis C	TB	HIV/AIDS	STIs	COPD	Cancer	Heart Disease	Diabetes	None
Age (in years)	1.01 0.99–1.02	1.02 1.00–1.04	0.98 0.94–1.02	1.02 1.00–1.03	0.99 0.98–1.01	1.06 1.04–1.08	1.07 1.05–1.09	1.08 1.05–1.10	1.05 1.04–1.06	0.96 0.96–0.97
Education (ref= less than high school)										
<i>High school diploma</i>	0.65 0.19–2.29	1.12 0.41–3.01	0.66 0.07–6.58	0.27 0.07–1.09	0.70 0.29–1.72	0.45 0.21–0.94	0.89 0.30–2.68	0.91 0.31–2.69	0.93 0.48–1.78	1.96 1.15–3.34
<i>Some college</i>	0.67 0.21–2.21	0.71 0.27–1.85	0.34 0.04–3.17	0.37 0.11–1.22	0.78 0.34–1.79	0.48 0.26–0.91	0.91 0.32–2.56	0.94 0.34–2.62	0.90 0.49–1.65	1.96 1.19–3.24
<i>Bachelor's degree or higher</i>	1.01 0.30–3.42	0.24 0.08–0.74	0.76 0.09–6.51	0.12 0.03–0.46	0.91 0.39–2.11	0.37 0.18–0.75	1.02 0.35–2.97	0.88 0.31–2.49	0.50 0.26–0.96	2.42 1.44–4.07
Race/ethnicity (ref= White, non-Hispanic)										
<i>Black, non-Hispanic</i>	1.45 0.53–3.96	1.65 0.71–3.86	5.19 0.81–33.34	2.14 0.71–6.43	1.24 0.70–2.19	0.40 0.18–0.90	0.83 0.35–1.98	0.77 0.29–2.02	1.60 1.00–2.54	0.75 0.51–1.12
<i>Other, non-Hispanic</i>	2.17 0.80–5.88	2.27 0.81–6.42	4.13 0.57–29.78	6.01 1.51–23.92	1.17 0.39–3.52	1.44 0.49–4.24	0.37 0.07–1.89	0.82 0.28–2.41	0.56 0.23–1.35	0.62 0.32–1.20
<i>Hispanic</i>	1.04 0.42–2.59	1.69 0.73–3.90	3.78 0.70–20.33	1.10 0.33–3.60	0.55 0.24–1.24	0.28 0.12–0.66	0.58 0.25–1.37	0.75 0.36–1.60	1.26 0.76–2.08	1.09 0.73–1.62
<i>2+ races, non-Hispanic</i>	0.36 0.08–1.66	5.39 1.40–20.77	-	-	1.13 0.47–2.73	3.10 0.82–11.71	0.15 0.03–0.62	0.46 0.12–1.72	1.28 0.33–4.99	0.77 0.36–1.64
Female sex	0.28 0.16–0.51	0.51 0.29–0.88	0.73 0.17–3.03	0.31 0.08–1.14	2.15 1.38–3.35	1.11 0.71–1.73	0.60 0.38–0.95	0.30 0.19–0.48	0.68 0.49–0.96	1.27 0.99–1.64
Married or living with partner	0.66 0.36–1.20	0.33 0.18–0.61	2.52 0.63–10.10	0.47 0.19–1.17	0.78 0.51–1.19	0.67 0.42–1.06	1.38 0.86–2.19	0.80 0.50–1.28	1.05 0.75–1.47	1.36 1.06–1.76
Household income 50,000USD or greater	0.69 0.39–1.23	0.40 0.21–0.75	1.10 0.30–4.06	1.22 0.50–2.99	1.01 0.65–1.56	0.46 0.28–0.76	1.22 0.77–1.94	0.76 0.49–1.19	0.70 0.50–0.98	1.26 0.98–1.63
Employed	0.42 0.22–0.79	0.32 0.17–0.58	0.58 0.15–2.16	0.36 0.14–0.94	1.25 0.80–1.94	0.17 0.11–0.29	0.41 0.25–0.67	0.32 0.20–0.50	0.40 0.29–0.57	2.56 1.97–3.32

Notes. TB= Tuberculosis, HIV/AIDS= Human immunodeficiency virus / acquired immune deficiency syndrome, STIs= Sexually transmitted infections, COPD= Chronic obstructive pulmonary disease. Statistically significant statistics are bolded.

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 4.

Clinical correlates of physical disease, odds ratios with 95% confidence intervals

	Alcohol-related liver disease		Hepatitis C	TB	HIV/AIDS	STIs	COPD	Cancer	Heart Disease	Diabetes	None
Age (in years) at regular use of primary substance	1.02 0.98–1.07		1.01 0.97–1.06	0.99 0.86–1.14	1.00 0.91–1.10	1.01 0.99–1.04	1.05 1.03–1.08	1.03 1.00–1.06	1.03 1.00–1.06	1.02 1.00–1.05	0.97 0.95–0.99
Number of substances used 10+ times	0.94 0.78–1.13		1.23 1.13–1.34	1.01 0.80–1.288	1.21 1.09–1.34	1.13 1.06–1.21	1.07 1.00–1.14	0.96 0.87–1.06	0.91 0.82–1.02	0.98 0.92–1.04	0.95 0.90–1.00
Years since problem resolution	0.97 0.95–1.00		1.01 0.99–1.03	0.96 0.90–1.03	1.02 0.99–1.04	0.99 0.98–1.01	1.03 1.01–1.04	1.05 1.03–1.06	1.04 1.03–1.06	1.03 1.02–1.04	0.98 0.97–0.99
Years smoked (lifetime)	1.01 0.99, 1.03		1.04 1.02, 1.05	0.98 0.92, 1.05	1.03 1.00, 1.06	0.99 0.98, 1.01	1.07 1.05, 1.09	1.02 1.00, 1.04	1.04 1.02, 1.06	1.04 1.03, 1.05	0.96 0.95, 0.97
Primary substance (ref=alcohol)											
<i>Cannabis</i>	0.41 0.08–2.22		0.90 0.20–4.13	5.35 0.93–30.74	1.21 0.15–9.89	1.69 0.84–3.39	0.67 0.21–2.13	0.44 0.18–1.11	0.57 0.21–1.56	0.42 0.21–0.85	1.22 0.74–2.02
<i>Opioids</i>	0.13 0.04–0.41		3.51 1.32–9.323.66	-	1.53 0.24–9.61	1.70 0.55–5.20	1.45 0.64–3.25	0.57 0.21–1.55	0.16 0.04–0.61	0.74 0.35–1.59	1.00 0.54–1.85
<i>Stimulants</i>	0.13 0.13–0.98		3.66 1.90–7.05	3.47 0.84–14.34	2.83 1.02–7.86	2.39 1.42–4.02	1.24 0.69–2.25	0.52 0.27–1.03	0.65 0.34–1.26	1.17 0.75–1.84	0.75 0.53–1.06
<i>Other drugs</i>	-		-	-	-	2.650.52–13.52	0.180.02–1.49	2.070.43–10.06	0.520.13–2.05	0.500.10–2.35	1.340.38–4.77
Outpatient addiction treatment	2.05 1.05–4.01		2.93 1.58–5.44	2.19 0.60–8.04	1.90 0.74–4.89	1.99 1.19–3.35	1.86 1.09–3.17	0.72 0.31–1.67	1.03 0.48–2.20	1.38 0.91–2.11	0.66 0.47–0.92
Inpatient addiction treatment	1.65 0.87–3.14		3.15 1.69–5.87	1.80 0.45–7.20	1.62 0.66–3.99	1.12 0.67–1.88	3.13 1.91–5.12	0.83 0.48–1.44	1.32 0.73–2.38	1.79 1.19–2.70	0.63 0.45–0.88
Mutual-help attendance (ref= never)											
<i>Former</i>	1.26 0.63–2.55		5.85 3.11–10.98	1.88 0.40–8.89	1.88 0.70–5.06	1.99 1.24–3.18	1.95 1.20–3.18	0.95 0.57–1.59	1.14 0.70–1.86	1.49 1.04–2.15	0.60 0.45–0.79
<i>Past 3-Months</i>	3.70 1.77–7.72		8.21 3.58–18.82	5.41 0.94–31.06	2.25 0.61–8.35	2.41 1.27–4.57	1.34 0.68–2.64	0.79 0.38–1.64	0.80 0.35–1.83	1.04 0.62–1.75	0.44 0.29–0.66
Quality of life	0.95 0.91–0.99		0.97 0.93–1.02	0.94 0.86–1.03	0.99 0.90–1.08	0.99 0.96–1.01	0.93 0.91–0.96	1.04 1.01–1.08	0.98 0.94–1.01	0.96 0.94–0.98	1.04 1.02–1.06
Happiness	0.85 0.60–1.20		0.78 0.55–1.10	0.62 0.42–0.90	1.37 0.89–2.13	0.96 0.78–1.19	0.87 0.65–1.17	1.45 1.13–1.85	0.96 0.70–1.32	0.97 0.80–1.17	1.00 0.87–1.15
Self-esteem	0.99 0.77–1.28		1.07 0.80–1.44	1.14 0.81–1.61	1.15 0.87–1.52	0.87 0.75–1.02	0.92 0.76–1.12	1.41 1.10–1.81	1.07 0.84–1.36	0.96 0.84–1.11	1.02 0.91–1.14
Psychological distress	1.09 1.04–1.13		1.02 0.97–1.07	1.11	0.97 1.02–1.20	1.04 0.88–1.07	1.03 1.00–1.07	0.95 0.90–0.99	1.00 0.95–1.05	0.98 0.94–1.01	0.99 0.97–1.02
Recovery capital	0.98 0.96–1.01		0.98 0.95–1.02	0.97 0.93–1.02	0.99 0.95–1.04	1.00 0.98–1.02	0.99 0.98–1.01	1.05 1.02–1.07	1.00 0.97–1.03	1.00 0.99–1.02	1.00 0.99–1.02

Notes, TB= Tuberculosis, HIV/AIDS= Human immunodeficiency virus / acquired immune deficiency syndrome, STIs= Sexually transmitted infections, COPD= Chronic obstructive pulmonary disease. Statistically significant statistics are bolded.

Table 5.

Demographic and clinical correlates of disease burden, odds ratios with 95% confidence intervals

	Disease Burden	
	1 vs. 0 diseases	2+ vs. 0 diseases
Age (in years)	1.03 (1.02–1.04)	1.07 (1.05–1.08)
Education (ref= less than high school)		
<i>High school diploma</i>	0.45 (0.25–0.81)	0.85 (0.39–1.83)
<i>Some college</i>	0.47 (0.27–0.80)	0.77 (0.37–1.60)
<i>Bachelor's degree or higher</i>	0.40 (0.23–0.71)	0.56 (0.26–1.19)
Race/ethnicity (ref= White, non-Hispanic)		
<i>Black, non-Hispanic</i>	1.35 (0.87–2.10)	1.32 (0.73–2.37)
<i>Other, non-Hispanic</i>	1.99 (0.98–4.05)	0.94 (0.36–2.48)
<i>Hispanic</i>	1.13 (0.72–1.75)	0.53 (0.30–0.94)
<i>2+ races, non-Hispanic</i>	1.28 (0.61–2.70)	1.34 (0.30–5.91)
Female sex	0.94 (0.70–1.26)	0.49 (0.34–0.70)
Married or living with partner	0.73 (0.54–0.98)	0.80 (0.55–1.15)
Household income 50,000 USD or greater	0.90 (0.68–1.21)	0.58 (0.40–0.84)
Employed	0.47 (0.35–0.63)	0.24 (0.16–0.35)
Age (in years) at regular use of primary substance	1.02 (1.00–1.05)	1.04 (1.02–1.07)
Number of substances used 10+ times	1.04 (1.00–1.11)	1.07 (1.00–1.15)
Years since problem resolution	1.02 (1.00–1.03)	1.04 (1.02–1.05)
Years smoked (lifetime)	1.03 (1.03, 1.04)	1.06 (1.04, 1.07)
Primary substance (ref= alcohol)		
<i>Cannabis</i>	1.00 (0.58–1.74)	0.40 (0.19–0.86)
<i>Opioids</i>	1.08 (0.52–2.22)	0.85 (0.40–1.80)
<i>Stimulants</i>	1.28 (0.88–1.87)	1.30 (0.79–2.13)
<i>Other Drugs</i>	0.78 (0.20–2.96)	0.69 (0.09–5.18)
Outpatient addiction treatment	1.19 (0.81–1.76)	2.53 (1.60–3.98)
Inpatient addiction treatment	1.31 (0.89–1.92)	2.28 (1.47–3.54)
Mutual-help Attendance (ref= never)		
<i>Former</i>	1.34 (0.97–1.84)	2.68 (1.79–4.02)
<i>Past 3-Months</i>	2.26 (1.42–3.62)	2.44 (1.41–4.23)
Quality of life	0.97 (0.94–0.99)	0.96 (0.93–0.99)
Happiness	0.99 (0.84–1.16)	1.00 (0.81–1.24)
Self-esteem	0.95 (0.84–1.07)	1.05 (0.89–1.24)
Psychological distress	1.01 (0.98–1.04)	1.00 (0.97–1.04)
Recovery capital	0.99 (0.98–1.01)	1.01 (0.99–1.03)

Note. Statistically significant statistics are bolded.